

Homework 7- Due Date SUN May 11, 11:59 P.M.

Problem 1. Let $X_1, X_2, \dots, X_{81}$ be a random sample of size $n = 81$ from a distribution with p.d.f.

$$f(x) = \begin{cases} \frac{(6-x)^2}{72} & 0 < x < 6 \\ 0 & \text{otherwise} \end{cases}$$

Find $P(\bar{X} < 1.6)$ approximately.

Problem 2. Use the Central limit theorem to approximate the following probabilities. Don't forget to apply the continuity correction (only if necessary).

- (a) Suppose we roll a fair 4-sided die until we get 100 threes. What is the probability it takes at least 1000 rolls until this happens?
- (b) Let X be the sum of 7500 real numbers, and Y be the same sum, but with each number rounded to the nearest integer before summing. If the fractions rounded off are independent and each one is uniformly distributed over $(-0.5, 0.5)$, use the Central Limit Theorem to estimate the probability that $P(|X - Y| > 50)$.

Problem 3. Let $X_1, \dots, X_n$ be a random sample from

$$f_{\theta}(x) = \begin{cases} (\theta + 1)x^{\theta} & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the maximum likelihood estimator (MLE) for θ .

Problem 4. Suppose we have a population which has a geometric distribution with parameter p . Take samples $X_1, \dots, X_n$ from this distribution.

- (a) Find the maximum likelihood estimation for p .
- (b) Find the method of moments estimation for p .